Curriculum Vitae

Xun Lin 林汛

Assistant Professor in Mathematics

linxun@cuhk.edu.cn | lin-x18@tsinghua.org.cn

Affil The Chinese University of Hong Kong, Shenzhen (CUHK–Shenzhen)

Office Room 410, 志仁楼, Longgang District, Shenzhen, Guangdong, China

linlinsai.github.io | Google Scholar

Education

September 2014 – June 2018 — **Bachelor in Mathematics, Southern University of Science and Technology, Shenzhen.** Supervisor: Qin Li.

September 2018 – June 2023 — **Ph.D. in Mathematics, Yau Mathematical Sciences Center, Tsinghua University, Beijing.** Thesis title: *Non-commutative aspects of Hodge conjecture, minimality, and Torelli problems*. Supervisor: *William Donovan*.

Research Experience

Sep. 2023 – Sep. 2025 — **Postdoctoral Researcher, Max Planck Institute for Mathematics, Bonn.** Mentor: Daniel Huybrechts.

Sep. 2025 – Sep. 2026 — **Postdoctoral Fellow in Mathematics, Hong Kong University of Science and Technology.** Mentor: Qingyuan Jiang.

Sep. 2026 – present — **Assistant Professor in Mathematics, The Chinese University of Hong Kong, Shenzhen.** Joined 1 September 2026.

Research

Algebraic geometry; geometry of derived categories; the noncommutative Hodge conjecture; GLSM models; Hochschild–Serre algebras of smooth proper dg categories; Hodge theory of algebraic varieties; Hitchin fibrations; categorical Torelli problems.

Preprints

10 Categorical Torelli theorem for smooth Fano varieties. With Shizhuo Zhang. arXiv:2610.25428 [math.AG].

9 Categorical Torelli theorem for (weighted) hypersurfaces via singularity category. With Jieheng Zeng and Shizhuo Zhang. arXiv:2610.00130 [math.AG].

8 Orlov's rationality conjecture for surfaces. With Shizhuo Zhang. arXiv:2609.21807 [math.AG].

7 Categorical reconstruction of del Pezzo surfaces: Hochschild–Serre algebras and spinor modifications. With Marco Rampazzo and Shizhuo Zhang. arXiv:2609.10344 [math.AG].

6 Group action of Hochschild-Serre algebra and categorical reconstruction. arXiv:2609.07562 [math.AG].

5 A note on the noncommutative Hodge conjecture for graded matrix factorizations. With Shizhuo Zhang. arXiv:2609.03784 [math.AG].

4 Semiorthogonal indecomposability for Hilbert schemes of points on integral locally planar curves. With Qingyuan Jiang. arXiv:2608.11608 [math.AG].

3 Infinitesimal Torelli problems for special Gushel–Mukai and related Fano threefolds: Hodge theoretical and categorical perspectives. With Shizhuo Zhang and Zheng Zhang. arXiv:2507.06995 [math.AG].

Research (continued)

2 Three approaches to categorical Torelli theorem for cubic threefold of non-Eckardt type via equivariant Kuznetsov components. With Xianyu Hu, Sebastian Casalaina-Martin, Shizhuo Zhang, and Zheng Zhang. arXiv:2405.20554 [math.AG].

1 Fully faithful functors, skyscraper sheaves, and birational equivalence. With Chunyi Li and Xiaolei Zhao. arXiv:2402.12573 [math.AG].

Papers

8 Categorical Torelli theorems for weighted hypersurfaces. With Jørgen Vold Rennemo and Shizhuo Zhang. *Advances in Mathematics* 503, Part C (2026), article 111271. arXiv:2408.08266.

7 On nonexistence of semi-orthogonal decompositions in algebraic geometry. *International Mathematics Research Notices*, 2026, Issue 2.

6 Serre algebra, matrix factorization and categorical Torelli theorem for hypersurfaces. With Shizhuo Zhang. *Mathematische Annalen* 391 (2025), 163–177.

5 Kuznetsov's Fano threefold conjecture via Hochschild–Serre algebra. With Shizhuo Zhang. *Mathematische Zeitschrift* 308 (2024), article 26. arXiv:2311.06450.

4 Categorical Torelli theorems for Gushel–Mukai threefolds. With Augustinas Jacovskis, Zhiyu Liu, and Shizhuo Zhang. *Journal of the London Mathematical Society* 109 (2024), e12878.

3 Indecomposability of the bounded derived categories of Brill–Noether varieties. With Chenglong Yu. *Proceedings of the American Mathematical Society* 152 (2024), 1457–1465.

2 Infinitesimal categorical Torelli theorems for Fano threefolds. With Augustinas Jacovskis, Zhiyu Liu, and Shizhuo Zhang. *Journal of Pure and Applied Algebra* 227 (2023), no. 12, 107418.

1 Noncommutative Hodge conjecture. *Journal of Noncommutative Geometry* 17 (2023), 1367–1389.

Works in Progress

3 Higgs objects over noncommutative varieties. With Shizhuo Zhang.

2 Categorical Torelli theorem for Verra threefolds and fourfolds. With Zhiyu Liu and Shizhuo Zhang.

1 Singular categories: Hodge theory and (birational) categorical Torelli theorem for nodal Fano threefolds. With Fei Xie, Daniele Faenzi, and Shizhuo Zhang.

Notes

Some remarks on Hochschild homology and semiorthogonal decompositions. Expository research note. arXiv:2112.10312 [math.AG].

Honors

Awards

2021 **Ruolin Prize for Graduate Students, Yau Mathematical Sciences Center.** For research excellence.

Scholarships

2018–2023 **Basic Scholarship for Graduate Students, Tsinghua University.**

2018–2022 **YMSC Scholarship, Yau Mathematical Sciences Center.**

Talks

October 2025 *Infinitesimal Torelli for special Gushel–Mukai threefolds*. Algebraic Geometry Seminar, Sun Yat-sen University, Guangzhou, China.

November 28, 2024 *IVHS via Kuznetsov components and categorical Torelli theorems for weighted hypersurfaces*. Seminar on Algebraic Geometry, University of Bonn, Germany.

Talks (continued)

September 26, 2024 — *IVHS via Kuznetsov components and categorical Torelli theorems for weighted hypersurfaces*. Algebraic Geometry Day Seminar, Morningside Center of Mathematics, Chinese Academy of Sciences.

July 2024 — *Categorical Torelli for smooth hypersurfaces*. Algebraic Geometry Seminar, Southern University of Science and Technology, Shenzhen, China.

February 2024 — *Infinitesimal categorical Torelli: an application to Kuznetsov's Fano threefold conjecture*. Oberseminar, Max Planck Institute for Mathematics, Bonn, Germany.

September 2023 — *A generalized Bondal–Orlov reconstruction theorem for fully faithful embeddings*. Five-minute talk, Junior Trimester Program, Hausdorff Center for Mathematics, Bonn, Germany.

April 2023 — *A new proof of Torelli for cubic fourfolds*. Seminar on Advanced Topics in Algebra and Geometry, Tsinghua University, Beijing, China.

March 2023 — *Infinitesimal categorical Torelli*. Geometric Representation Theory Seminar, Tsinghua University, Beijing, China.

November 2022 — *Hodge theory of combinatorics: an introduction*. Tsinghua Preprint Seminar, Tsinghua University, Beijing, China.

October 2022 — *Categorical Torelli*. Tsinghua Preprint Seminar, Tsinghua University, Beijing, China.

July 2022 — *Indecomposability of the bounded derived categories of Brill–Noether varieties*. Online poster session, VBAC Seminar, University of Warwick.

April 2022 — *Nonexistence of semiorthogonal decompositions and sections of paracanonical systems*. Algebraic Geometry Seminar, Stony Brook University, USA.

March 2022 — *Nonexistence of semiorthogonal decompositions and sections of paracanonical systems*. Algebraic Geometry Seminar, Cornell University, USA.

January 2022 — *Nonexistence of semiorthogonal decompositions and sections of paracanonical systems*. Geometry and Topology Seminar, Southern University of Science and Technology, Shenzhen, China.

Fall 2021 — *VGIT and derived categories*. Student Seminar, Tsinghua University, Beijing, China.

September 2021 — *Nonexistence of semiorthogonal decompositions and the base locus of paracanonical systems*. Tata Institute of Fundamental Research, India.

June 2021 — *Derived categories of symmetric products of curves*. Internal Seminar, University of Edinburgh, UK.

May 2021 — *Noncommutative Hodge conjecture*. Geometric Representation Theory Seminar, Tsinghua University, Beijing, China.

March 2021 — *Noncommutative Hodge conjecture*. Southern University of Science and Technology, Shenzhen, China.

Noncommutative Hodge conjecture. ShanghaiTech University, Shanghai, China.

Fall 2019 — *Examples of Shimura varieties*. Student Algebraic Geometry Seminar, Tsinghua University, Beijing, China.

Spring 2019 — *Derived Torelli for K3 surfaces*. Student Algebraic Geometry Seminar, Tsinghua University, Beijing, China.

Nonexistence of semiorthogonal decompositions and sections of the canonical bundle. Student Algebraic Geometry Seminar, Tsinghua University, Beijing, China.

Fall 2018 — *Geometry of moduli spaces of sheaves on surfaces*. Student Algebraic Geometry Seminar, Tsinghua University, Beijing, China.

Teaching

Instructor

2026–2027 Spring — **MATH 3005: Abstract Algebra II.** The Chinese University of Hong Kong, Shenzhen. Scheduled. Lecture notes: *Commutative Algebra and Basic Algebraic Geometry*, Version 7, October 2026.

2025–2026 Spring — **Foundation of Mathematics.** Hong Kong University of Science and Technology. Course notes.

TA

- General Topology, TA. Tsinghua University.
- Mathematical Analysis II, TA. Tsinghua University.
- Functional Analysis I, TA. Tsinghua University.
- Calculus A, TA. Tsinghua University.
- Algebraic Geometry I, TA. Tsinghua University.

2022 — Algebra I, TA. Qiuzhen College, Tsinghua University.

Seminars

2027 (planned) — **Nonabelian Gauge Linear Sigma Model.** Organizer: Xun Lin. Dates, venue, and programme to be announced. Seminar information.

Fall 2022 — **Preprint Seminar.** Student seminar at Tsinghua University. Seminar website.

Visiting and conferences/workshops

April 20–25, 2026 — *Workshop on Moduli Spaces in Various Flavors of Geometry*. Kyoto University, Kyoto, Japan.

December 10–15, 2025 — *Workshop on Derived Categories*. Yonsei University, Seoul, Korea.

July 7–11, 2025 — *Derived Categories, Noncommutative Geometry and Deformation Theory*. Cetraro, Italy.

March 24–28, 2025 — Technical University of Munich, Munich.

October 13–18, 2024 — Sapienza Università di Roma, Rome.

July 15–22, 2024 — Southern University of Science and Technology, Shenzhen, China.

June 27–September 30, 2024 — Morningside Center of Mathematics, Chinese Academy of Sciences, Beijing, China.

June 26–July 24, 2024 — Yau Mathematical Sciences Center, Tsinghua University, Beijing, China.

June 4–22, 2024 — Shanghai Center for Mathematical Sciences, Fudan University, Shanghai, China.

May 26–31, 2024 — Stockholm University, Sweden.

January 22–26, 2024 — University of Genoa, Italy.

December 15–20, 2023 — University of Oslo, Norway.

September–December 2023 — *Junior Trimester Program*. Hausdorff Center for Mathematics, Bonn, Germany.

Visiting and conferences/workshops (continued)

July 26–August 10, 2023	Sichuan University, Chengdu, China.
July 10–21, 2023	Tsinghua University, Beijing, China.
March–June 2022	Cornell University, USA.
March 2021	ShanghaiTech University, Shanghai, China.
September 2019	*Geometry of Derived Categories*. University of Liverpool, UK.
May 2019	*Derived Categories Seminar*. Sun Yat-sen University, Guangzhou, China.
	Algebraic Geometry for Young Scholars. University of Science and Technology of China, Hefei, China.

Languages

Native

- Cantonese and Mandarin Chinese.

Advanced

- English.

Updated 30 October 2026.